James O'Connell

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Experience

Software Engineer Manager, Raytheon (TS/SCI) **Dec. 2023 – Present**
Python, TypeScript, Java, C++, Docker, Kubernetes, Gradle *Denver, CO*

- Developed and enhanced C2 ground systems supporting real-time telemetry processing, commanding and automation.
- Collaborated with spacecraft operators and product teams to align C2 system design with mission needs.
- Led a cross-functional software team providing technical guidance, staff management, career development, and coaching.
- Championed true Agile practices, focusing on users, rapid iteration, and empowering teams to improve delivery.
- Coordinated intern program, pairing developers with impactful projects for hands-on development and growth.
- Spearheaded low-to-high development processes, ensuring secure, compliant C2 delivery in classified environments.

Full-Stack Software Engineer, USAA **May 2021 – Dec. 2023**
Java, SpringBoot, Kafka, React.js Node.js, Docker, AWS, OCP, Hibernate/JPA, DB2 *Phoenix, AZ*

- Developed and maintained event-driven P&C Insurance Communications System Java APIs and Kafka streams.
- Reduced toil of O&M availability teams by building resources to reconcile reinstated communication packets.
- Identified & triaged on–call production issues/defects during and after code deployments.
- Developed features for interns and guided the them on Java API development, Agile methodologies and SDLC.

Teacher, Microsoft TEALS **Oct. 2020 – May 2021**
Mentorship, Education, Python, CS Foundations *Tucson, AZ*

- Educated high school students in computational thinking, problem solving, coding, and computer science concepts.
- Provided a hands–on learning in which students' learn through discovery, experimentation, and application.

Software Engineer, Lunar & Planetary Laboratory **May 2019 – Jan. 2020**
Python, C, Remote Sensing, Simulation, Analysis, Research *Tucson, AZ*

- Developed/maintained simulation software utilizing air/spaceborn sounding radargrams to assist planetary investigation.
- Collaborated in the analysis of remote sensing data to better understand debris covered glaciers on Mars and Earth.
- Provided requirements analysis to shape the roadmap to the needs of research specialists.

Infantry Fireteam Leader, U.S. Army, 82nd Airborne Division **Jan. 2015 – May 2018**
Leadership, Employee Development, Resource Management, Risk Analysis, Planning *Fort Bragg, NC*

- Executed the planning/assessment of training exercises in high pressure, fast moving, dynamic and ambiguous scenarios.
- Provided 1:1 performance and growth development counseling to enhance ones current and future responsibilities.
- Led 2–4 man fireteams in airborne/combat operations across austere environments(Afghanistan, Jun. 2017 – Mar. 2018).

Projects

Production Support Tool, *Java, SpringBoot, React.js, SQL, Docker, AWS/OCP* **Jun. 2023**

- Designed and built Java APIs with a React web app interface that provides accessibility, and streamlines manual tasks.
- Implemented an event listener to provide oversight and accountability on users executing data modifications.
- Leverages Hazelcast for distributed in-memory caching to improve latency, flexibility, and manageability.

Surface Clutter Simulator, *Python, C, numpy, gdal, cTypes* **Jan. 2020**

- Generates two-dimensional left/right-side(of the spacecraft) cluttergram images, each containing surface reflections.
- Leverages digital surface models, geographic, geometric and ionospheric properties sourced from NASA's PDS as inputs.
- Increases confidence that interpreted subsurface features in radargrams are not a product of surface topography.

Skills

Languages: Java, Python, TypeScript, JavaScript, C, C++, SQL, HTML, CSS
Frameworks: SpringBoot, React.js, Angular, JUnit, Jest, JPA/Hibernate
Databases & Messaging: PostgreSQL, Oracle DB, IBM DB2, Kafka, JMS
Cloud & DevOps: Docker, Kubernetes, AWS, OpenShift, Helm, Git, GitLab, Jenkins, Gradle

Education

Master of Science in Computer Science, *Georgia Institute of Technology* **May 2026**
Bachelor of Science in Computer Science, *University of Arizona* **Dec. 2021**

Publications

Christoffersen, M. S.; Holt, J. W.; Kempf, S. D.; O'Connell, J. D. *MRO SHARAD Clutter Simulations Data Products*. 2021. PDS Geosciences (GEO) Node. https://doi.org/10.17189/nbdh-2k53